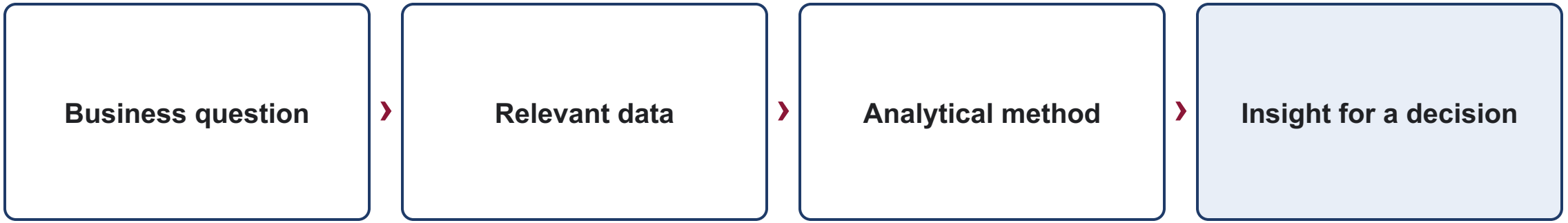
SEMINAR 4

Overview of Data Analytics

Which analytical technique fits which business question?

Business Question, Data, Analytics, Insight



Seminar 4 Learning Objectives

Match six techniques to business questions

Distinguish target-based and pattern-discovery tasks

Interpret outputs without claiming causality or certainty

Place technique choice in an analytics workflow

Churn: Business Concern and Retention Question

ILLUSTRATIVE EXAMPLE

A telecommunications provider wants to identify customers nearing renewal whose observed behavior may warrant a retention conversation.

This is a teaching example, not a company result.

Decomposing the Retention Problem

Risk question: Who may leave? › **Evidence question: What information supports the assessment?** › **Action question: Which retention action deserves evaluation?**

Risk Prediction Is Not Intervention Evaluation

Risk estimate

Intervention evaluation

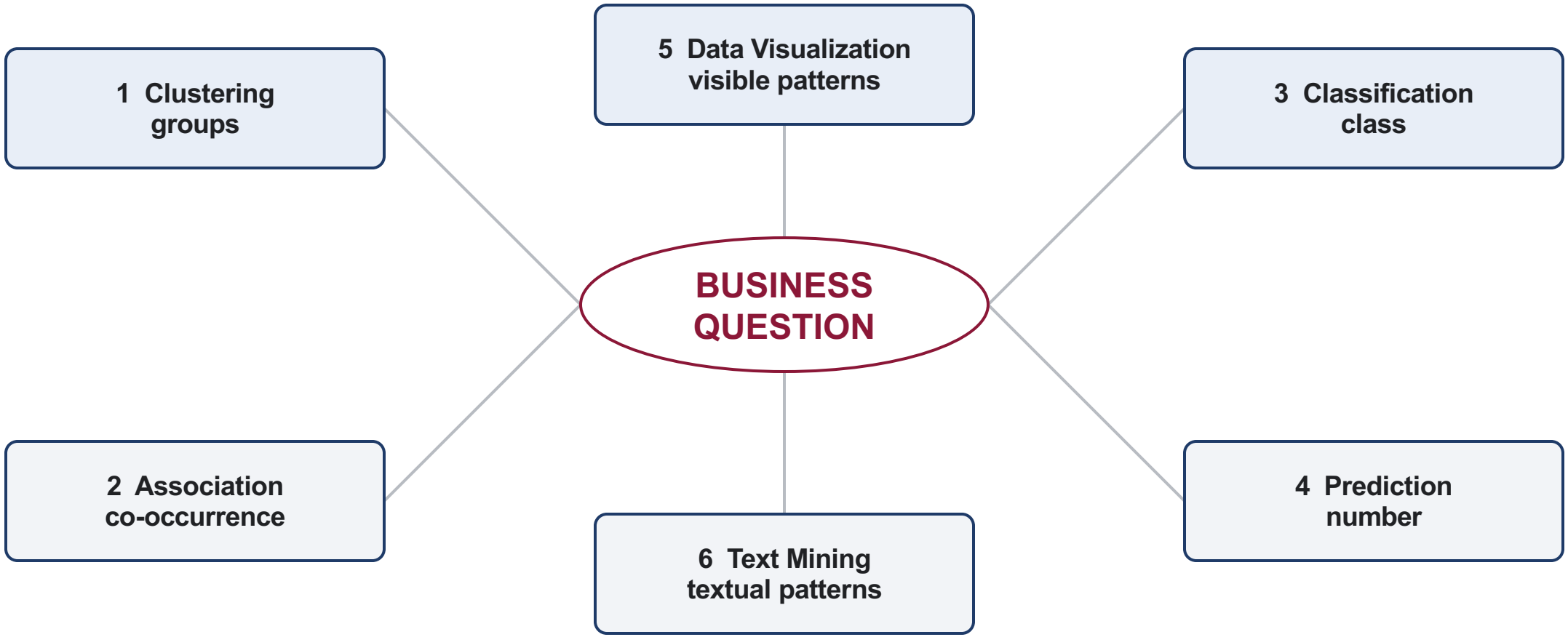
A prediction alone does not show that an offer caused retention.

Data-Mining Techniques

Different techniques answer different kinds of questions from data.

Technique choice follows the question and available information.

Six Technique Families



Clustering

Business question: Which customers or objects form useful groups?

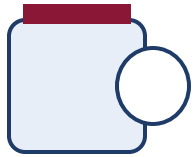
Output: group labels based on similarity

Limit: a group is not automatically a business category

Similarity as a Sorting Question

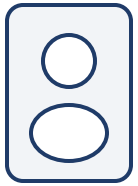
ILLUSTRATIVE ACTIVITY

How would you group these products?



Coffee cup

Beverage
Price: low
Buy: daily



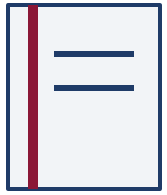
Smart speaker

Electronics
Price: high
Buy: occasional



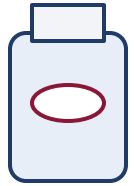
Running shoes

Apparel
Price: mid
Buy: seasonal



Notebook

Stationery
Price: low
Buy: monthly



Shampoo

Personal care
Price: mid
Buy: monthly

State the attribute you used.

More than one grouping can be reasonable.

Attributes for Grouping

PRICE RANGE

Low • Mid • High

PURCHASE FREQUENCY

Daily • Monthly • Occasional

PRODUCT CATEGORY

Beverage • Apparel • Electronics

Selected attributes shape the groups.

Category codes need interpretation:

numbers do not make categories naturally close.

Distance Depends on Representation

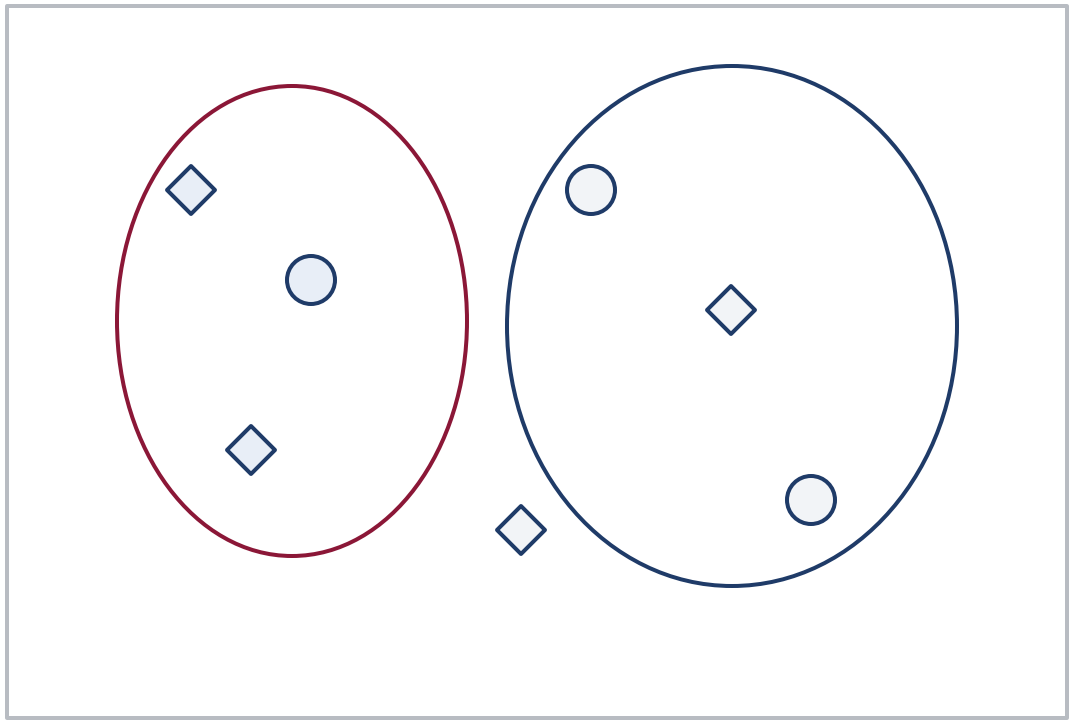
Different attributes or scales can change the comparison.



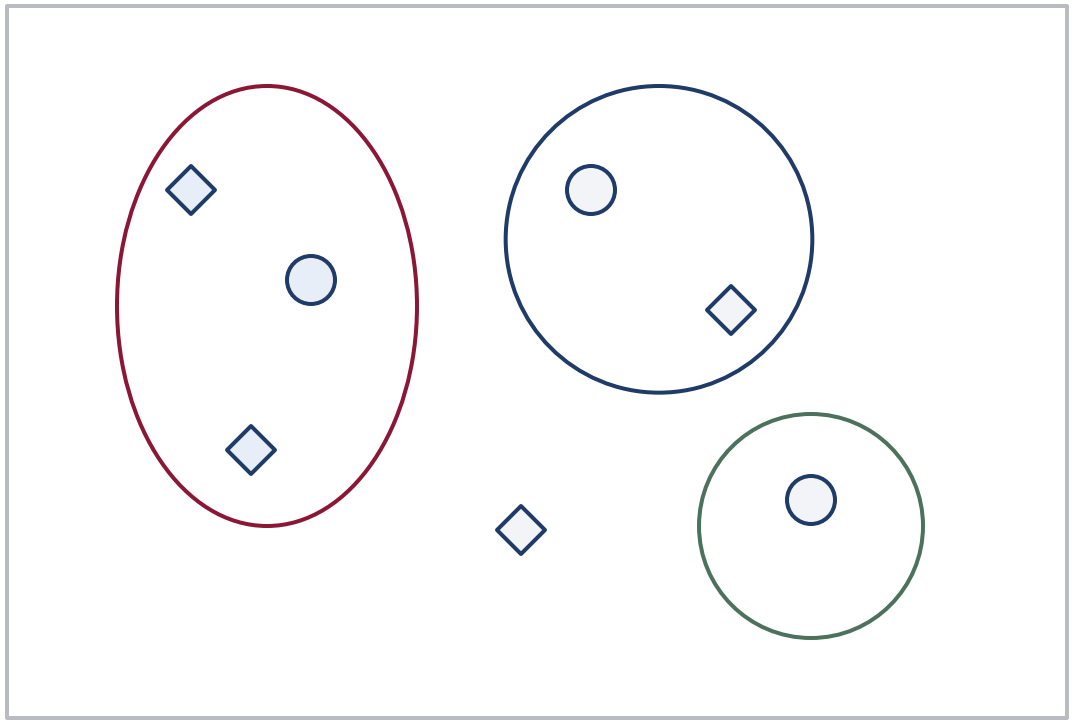
Distance is a modeling choice, not objective truth.

More Than One Grouping Can Be Useful

Two broad groups



Three specific groups



Different groupings can support different decisions.

Customer Segments as Decision Inputs

Possible use: tailor a retention message by observed segment

›

Check whether the segment is stable, understandable, and useful

›

A segment does not explain why customers behave that way

Association

Business question: What tends to occur together?

Output: a co-occurrence pattern

Limit: co-occurrence does not establish cause

Items Purchased Together

ILLUSTRATIVE EXAMPLE

A retailer examines which items tend to occur together in transaction records.

Use the pattern to ask a merchandising question, not to claim causality.

Classification

Past labeled cases

class A
class B

›

New case
CLASS LABEL

Classification: A Discrete Target

ILLUSTRATIVE EXAMPLE

Customer cases

renew
not renew

›

Likely class
RENEW / NOT RENEW

Prediction

Past numeric cases

inputs

›

New case
ESTIMATED NUMBER

Estimate, not certainty

Prediction: A Numeric Target

ILLUSTRATIVE EXAMPLE

Past demand patterns

history + context

›

Next-month demand
NUMERIC ESTIMATE

Estimate, not certainty

Historical Example: Behavioral Inference

HISTORICAL EXAMPLE — 2012 REPORTING

PURCHASE HISTORY

behavioral signals

↓

sensitive inference

Reporting described Target's use of purchase-history analytics to estimate pregnancy likelihood.

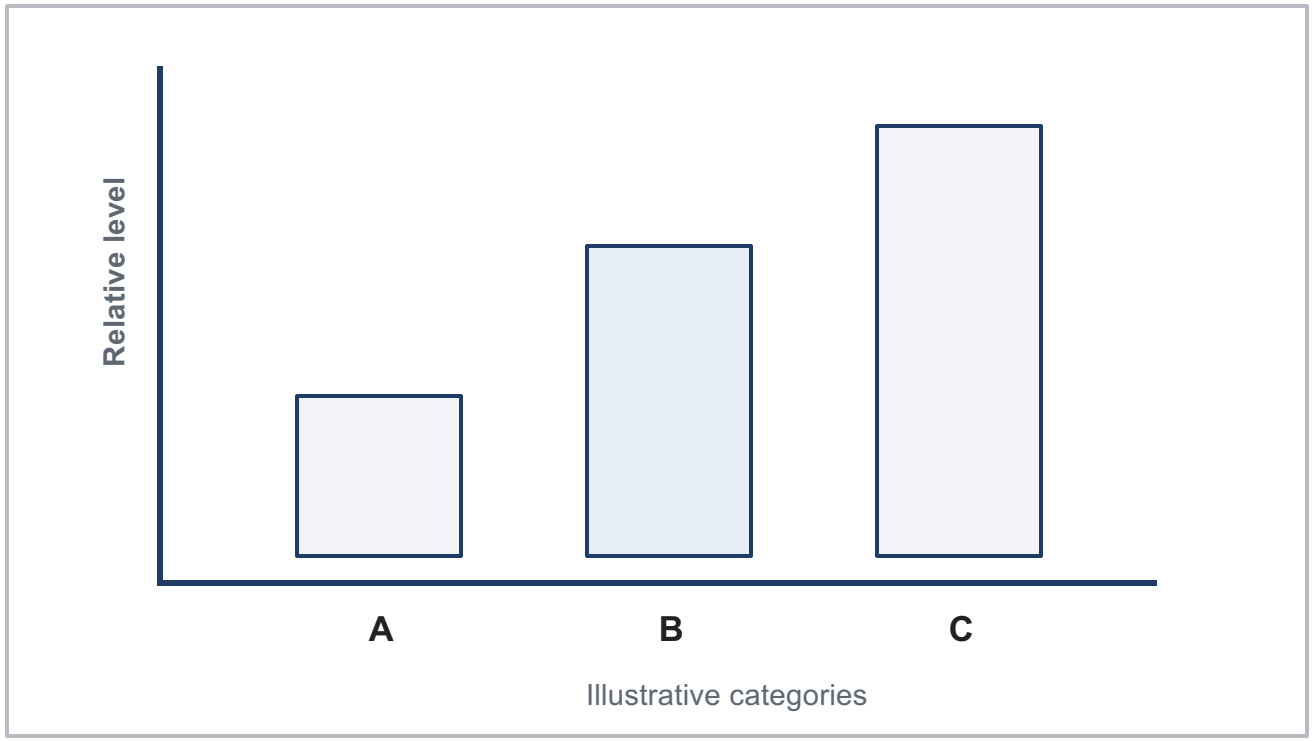
Teaching use: prediction, privacy, and limits of inference.

Do not treat a prediction as a fact about a person.

Source: Duhigg, The New York Times Magazine (2012)

Data Visualization

ILLUSTRATIVE / CONCEPTUAL



Question

What do plots reveal about distributions or relationships?

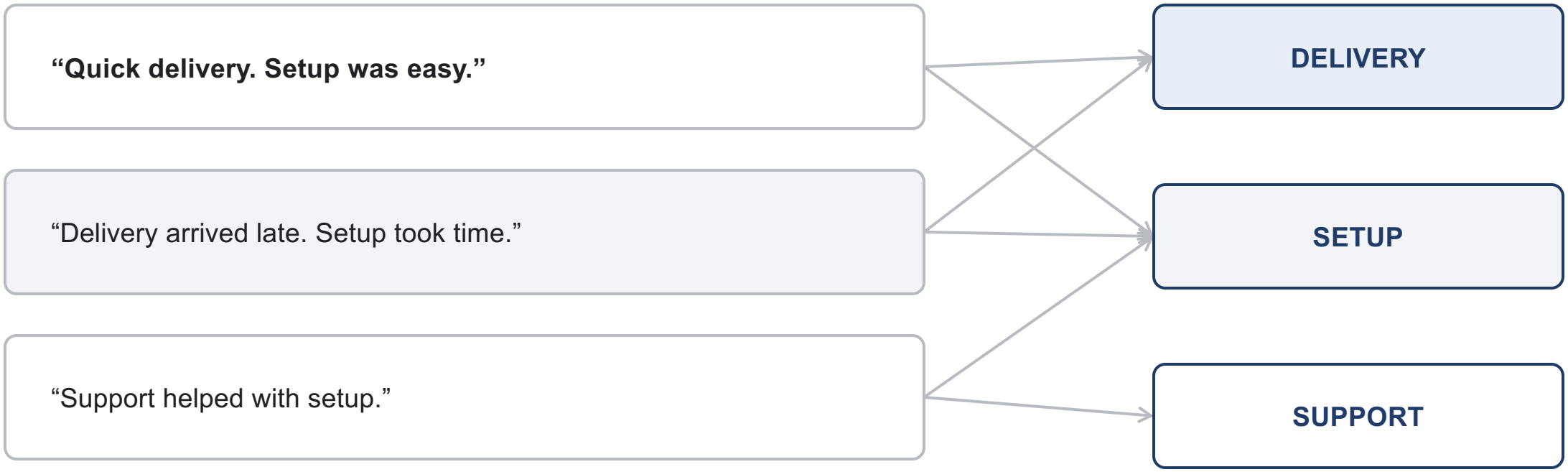
Output

Visible patterns

This is a concise preview before the focused visualization module.

Text Mining

FICTIONAL TEXT SAMPLE



Question: What patterns appear in text? Output: textual patterns

Preview only. No real corpus is implied.

Learned Patterns and Explicit Rules

Programs include explicit choices and rules.

Models can learn patterns from examples.

Human judgment remains part of the process.

Target Availability: Four Task Types

KNOWN TARGET

3 Classification
Output: class

4 Prediction
Output: number

NO KNOWN TARGET

1 Clustering
Output: groups

2 Association
Output: co-occurrence

Four task types are compared here. Visualization and Text Mining remain in the six-technique spine.

Supervised Tasks

A known target guides learning.

Classification predicts a class.

Prediction estimates a number.

Unsupervised Tasks

No known target defines the answer.

Clustering finds useful groups.

Association finds co-occurrence patterns.

Statistics, Querying, and Analytics

Statistics: summarize data and help quantify patterns and uncertainty.

Querying retrieves specified information.

Analytics and data mining discover or estimate patterns. These activities can overlap.

Technique Identification Discussion

ILLUSTRATIVE ACTIVITY

Identify the business question.

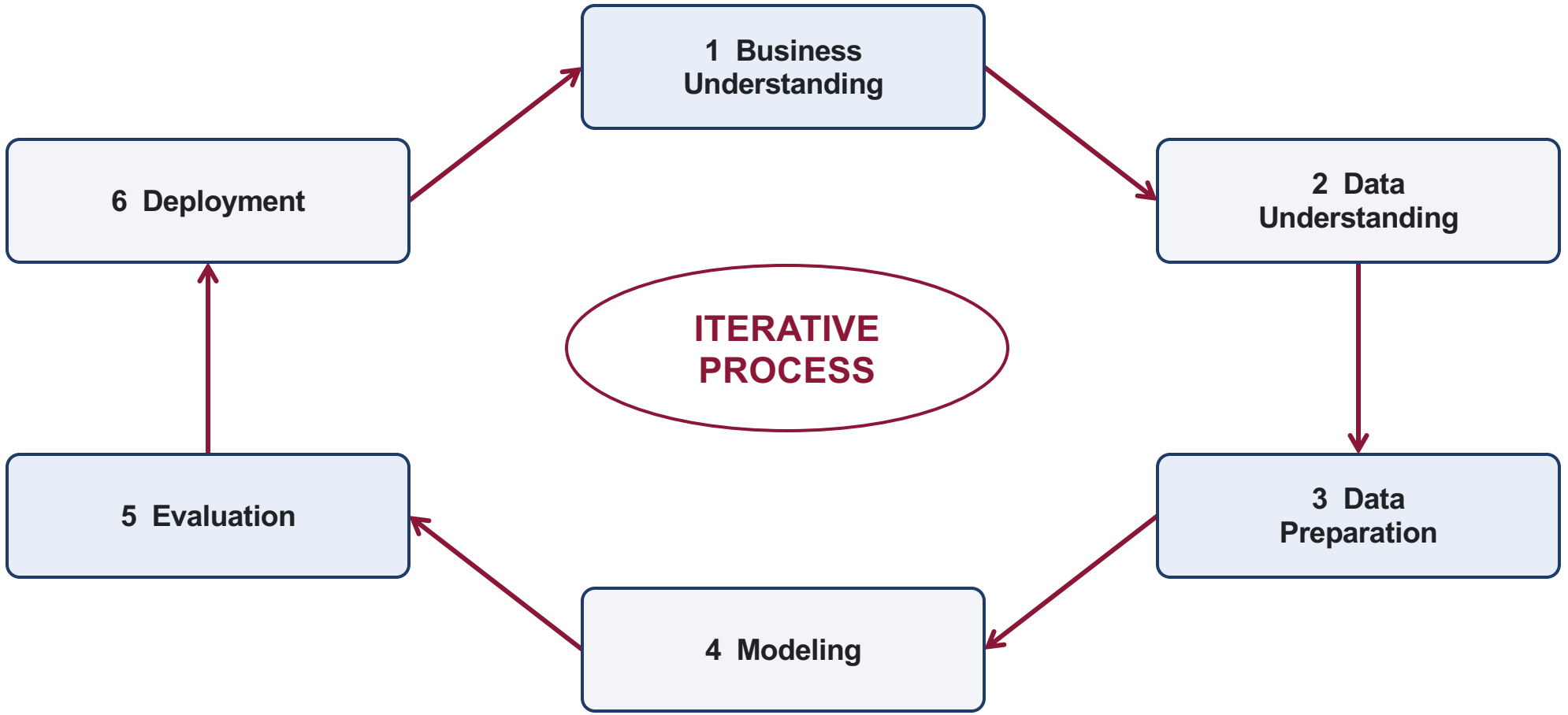
Name the conceptual output.

Select the technique that fits.

Six-Technique Comparison

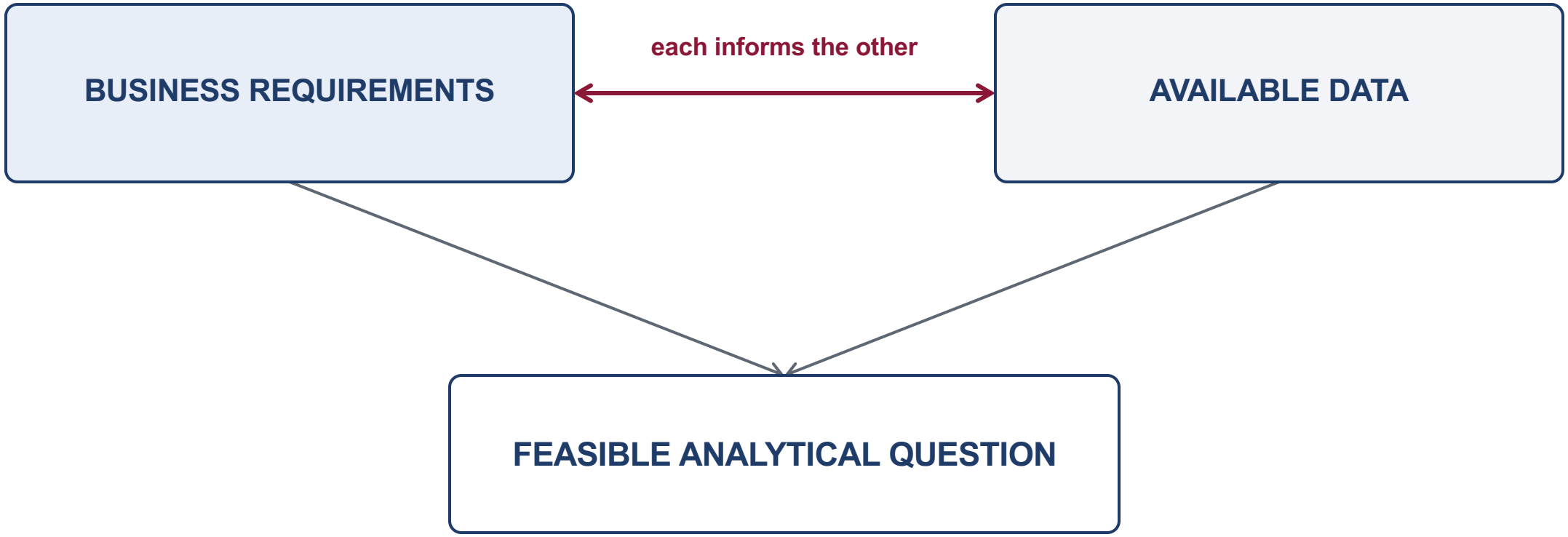
Technique	Typical question	Output
1 **Clustering**	Which cases form useful groups?	**groups**
2 **Association**	What tends to occur together?	**co-occurrence**
3 **Classification**	Which discrete class is likely?	**class**
4 **Prediction**	What numeric value is likely?	**number**
5 **Data Visualization**	What do plots reveal?	**visible patterns**
6 **Text Mining**	What patterns appear in text?	**textual patterns**

CRISP-DM 1.0



CRISP-DM 1.0 (Chapman et al., 2000)

Requirements and Available Data



Technique Selection and Modeling



Evaluation

Validate the result

Compare alternatives

Assess stakeholder consequences

Deployment Supports Decisions



Close: Ask, Match, Interpret

Ask a clear business question

Match a fitting technique

Interpret the result and its limits